Blockchain-Based Secure Online Voting System

Objective

To build a secure, transparent, tamper-proof online voting system using Blockchain that maintains voter anonymity, prevents duplicate voting, and ensures even super administrators cannot access confidential voting data.

Key Innovations

- Zero-Knowledge Proofs (ZKP) for voter eligibility without identity disclosure
- Blind Signatures to prevent voter-vote linkage
- Commit-Reveal voting mechanism to avoid vote tampering
- Decentralized Identity (DID) for privacy-preserving authentication
- Threshold Decryption where no single authority can decrypt votes
- Zero-Trust Admin Architecture

System Architecture

The system consists of a voter application, decentralized identity verifier, blockchain network, smart contracts, IPFS for encrypted vote storage, zero-knowledge proof engine, and distributed election committee key holders.

Voting Workflow

- Voter registers using verified identity to receive an anonymous voting token.
- Blind signature validates the token without revealing voter identity.
- Voter submits encrypted vote hash (commit phase).
- Zero-knowledge proof verifies eligibility and prevents duplicate voting.
- After voting ends, encrypted votes are revealed and verified.
- Votes are decrypted using threshold decryption.

Security Guarantees

- Immutable blockchain records prevent tampering
- Nullifier hashes stop duplicate voting
- Admins cannot view, modify, or link votes
- Public vote counting with private vote content

Why This Solution Wins Hackathons

This system combines real-world election requirements with advanced cryptography, offering trustless governance, high transparency, and unmatched privacy. It goes beyond typical blockchain voting demos by eliminating administrator trust entirely.